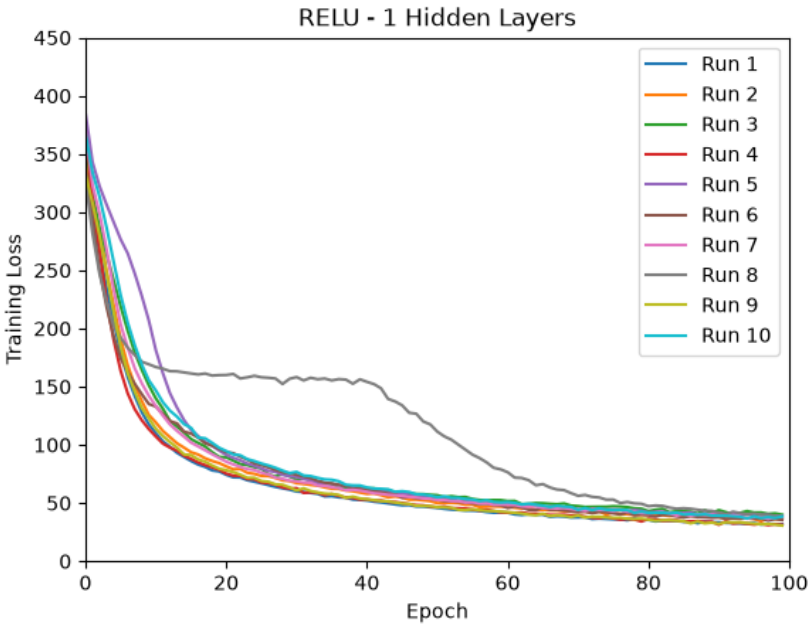
One hidden layer

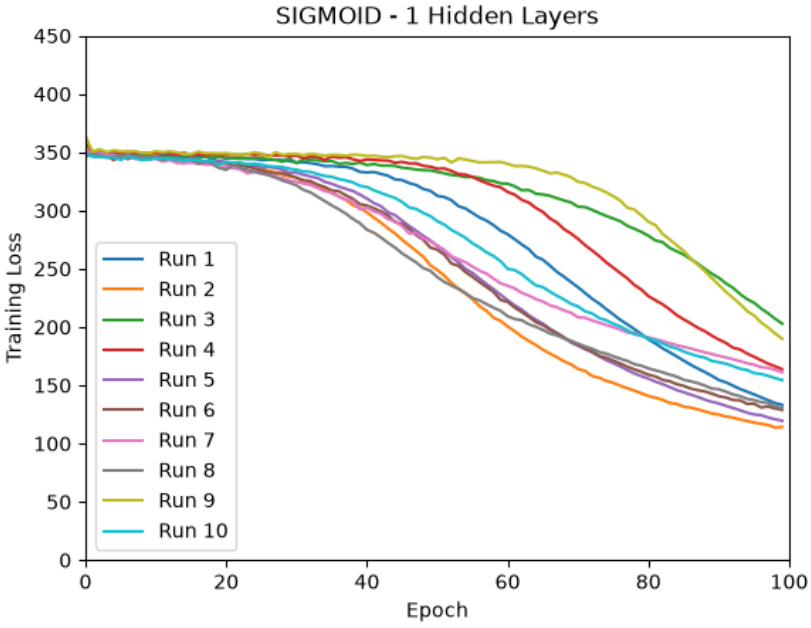
500 noisy XOR samples per run | 10 runs | 100 epochs | learning rate 0.01 | 10 units per layer

Figure 1 / ReLU - 1 hidden layer



Loss generally falls rapidly and approaches approximately 30-45 by epoch 100, with some variation between initializations.

Figure 2 / Sigmoid - 1 hidden layer

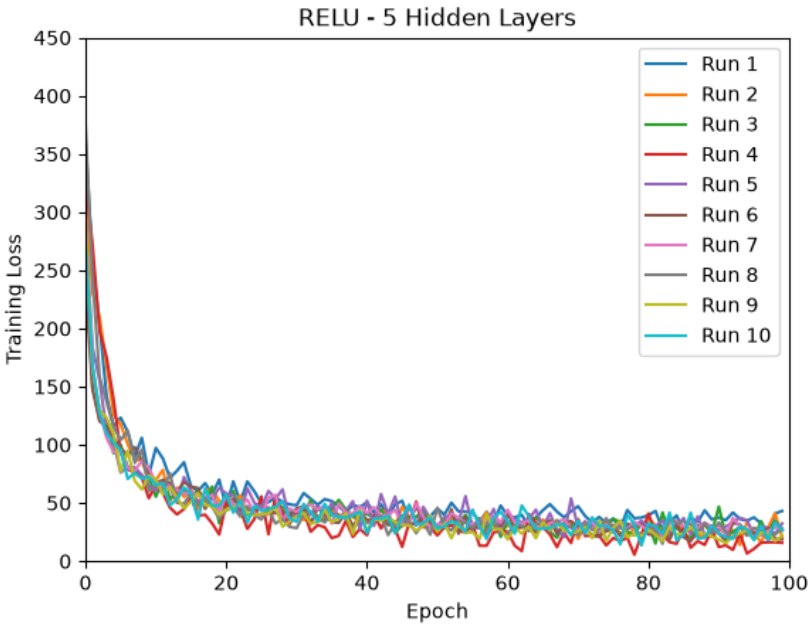


Loss decreases much more slowly, with substantial variation in the time each run begins to improve.

Five hidden layers

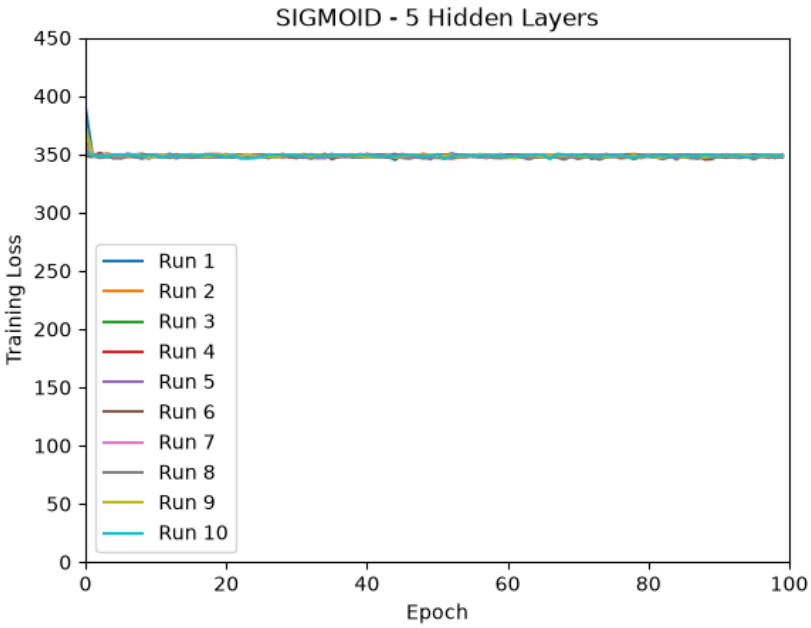
Same layer width, samples, epochs, and learning rate as the one-layer networks

Figure 3 / ReLU - 5 hidden layers



Loss falls quickly in all ten runs, despite some noisier epoch-to-epoch changes.

Figure 4 / Sigmoid - 5 hidden layers



All ten runs remain near a loss of 347, indicating little learning over 100 epochs.

Gradient measurements and interpretation

First-hidden-layer weight gradients during epoch 1 | 10 runs per configuration

Table 1. Mean absolute first-layer weight gradient

Activation	Hidden layers	Mean absolute gradient
Sigmoid	1	**0.02023466**
ReLU	1	**0.07806709**
Sigmoid	5	**0.00002485**
ReLU	5	**0.09106521**

How the gradients were measured

For each configuration, a newly initialized network was trained for one epoch on the same 500-sample noisy XOR dataset. The absolute derivatives of the first hidden layer weights were collected before each SGD weight update and averaged across weights, samples, and ten runs.

Interpretation

The five-layer sigmoid model has a first-layer gradient of approximately 0.00002485, about 814 times smaller than its one-layer counterpart (0.02023466). This is consistent with vanishing gradients: early-layer weights receive very small training signals, which helps explain the nearly flat loss curves in Figure 4.

ReLU maintains much larger first-layer gradients in both the one-layer and five-layer networks. The corresponding learning curves show that both ReLU configurations reduce training loss successfully. Together, these measurements support the conclusion that ReLU is better suited to training deeper networks under the tested conditions.

Experimental notes

Each learning-curve run used a newly sampled noisy XOR dataset and independently initialized weights; thus, some run-to-run differences reflect both sample variation and initialization. The separate gradient comparison used the same dataset across runs. Identical fixed data for all learning-curve models would provide a more strictly controlled comparison.

Conclusion

Increasing depth from one to five layers severely reduces first-layer gradients with sigmoid, but not with ReLU. The gradient measurements and training curves provide mutually consistent evidence of a vanishing-gradient problem.